The Origins of the Crossover Phenomenon

Louis-Harry Desouvrey

December 2025

First draft

louish8@gmail.com

Abstract

In this paper, I present a novel analysis of the the crossover phenomenon, which is based on the nonlinear theory of coreference as well as the special syntax of vectors. I show that the crossover effect is strong when either: (a) the dependency is modified under extraction of an anaphor, (b) a *wh*-vector anteceding a pronoun remains active, as opposed to being disabled, after extraction, (c) a permanent line-crossing effect appears in the derivation or (d) the structure contains multiple ill-formed states. On the other hand, weak crossover is due either to multiple transitory constraint violation in the derivation or to one ill-formed state in a multiple-state structure. In addition, it is shown that any dependency is weaken if the antecedent and the antecedee are not in the same plane. This analysis is extended to the pair-list reading, which appears to be mainly a consequence of the entanglement of the universal with every referential feature in its domain.

Keywords: strong crossover, weak crossover, referential features, functional features, plane, vector effects, quantifier entanglement, feature spinning, *wh*-operators.

1. Introduction

Generally speaking, the acceptability of a sentence is distorted when a referring element is in a higher position than another referring element it depends on, as in (1a-d). This state persists after *wh*-extraction of the antecedent to a higher position, as shown in (1e,f).[1] According to the judgment reported in the vast literature on this phenomenon (Agüero-Bautista 2012, Lasnik and Stowell 1991, Postal 1993, Ruys 2000, Safir 2015, etc.), sentences (1a,c,e) are worst than (1b,d,f), which justifies their labeling as strong crossover (SCO) and weak crossover (WCO) respectively.[2]

(1) a. **He* loves *Paul*.
b. *?*His* mother loves *Paul*.
c. **He* loves *everyone.*
d. *?*His* mother loves *everyone*.
e. *Who$_i$ does he$_i$ love t_i ?
f. *?Who$_i$ does his$_i$ mother love t_i ?

Current accounts of this phenomenon are mostly based on Binding theory as well as various related theoretical notions such as traces, *c*-command, abstract movement, LF, etc.[3] Those accounts rely also on formal logic, about which I have nothing to say. In this paper, I will propose a novel account of the

1 The classical sentences in (1) are found across the literature. I have no personal judgment on them whatsoever.

2 As pointed out by Safir (2015), the term crossover was coined by Postal (1971), who first investigated the phenomenon, and the distinction between SCO and WCO is due to Wasow (1972/1979).

3 In a computational approach, Shan and Barker (2006) posit that crossover effects obtain because expressions must be evaluated from left to right, which implies that the antecedent must precede any dependent. This generalization is correct, and the theory to be presented here will explain why it is so.

crossover phenomenon. Given the feature-based theory of coreference proposed in Desouvrey (2003), which draws on nonlinear phonology and morphology, I will show that the acceptability of (1e) is strongly distorted because it is derived from the transformation of an ill-formed structure. In the grammatical counterpart of (1a), the object may not be extracted in the first place. In effect, the operator incorrectly comes to replace an anaphor, and hence the dependency relation is reversed in the structure. On the contrary, the derivation of (1c,d,f) only violates various constraints on vectors and the prohibition of line crossing in the nonlinear representation.

I proceed as follows. In the next section, I present the theory of coreference proposed in Desouvrey (2003), and I introduce the notion of functional features, as opposed to referential features. In section 3, I briefly present the analysis of *wh*-extraction, which is already discussed in earlier work, in order to illustrate various vector effects. Then in section 4, I proceed to account for the crossover effects, discussing in turn *wh*-operators, the universal quantifier, and the case of pair-list reading. Before concluding the paper, I consider in section 5 various constructions, including relative clauses, parasitic gap, *tough*-movement, and topicalization, where weak crossover is absent, as discussed in the literature.

2. Coreference and features

Two types of features will be distinguished: referential features (RF), which materialize real world entities in the grammar, and functional features (FF), which interact with the former in the structure. Both types of features are dependent of the referential articulation, or *R*-node. The relation of coreference between the three types of referring element, namely NPs, pronouns, and anaphors, is simply realized by spreading a RF or a *R*-node to a dependent element.

2.1 Referential features

In the theory of coreference proposed in Desouvrey (2003), referring elements show a three-way distinction. The autonomous ones, for instance a proper noun like *Mary* and a common noun like *book*, bear a RF that is denoted by their first letter, [M] and [b], respectively. RFs depend on an intermediate node referred to as the *R*-node, which is itself originated from a root node.[4] The root node, which can be neutral or not, will be ignored here. A third person pronoun has a bare *R*-node, and therefore it must get a RF from an NP. First and second person pronouns are autonomous, as they are constantly specified for their own RF, represented by the numerals *1* and *2* respectively. On the other hand, an anaphor comes with a bare root node, and can only receive a *R*-node from an NP or a pronoun. Let us illustrate this with the following examples.

(2) a. Paul$_i$ likes him$_j$.
b. He likes Mary.

The NPs *Paul* and *Mary* are autonomous in the sentence and the discourse, as they bear their own RF, *P* and *M* respectively, as shown in (3). On the other hand, the pronouns *him* in (3a) and *he* (3b) are also free in their respective sentence, but they must be linked by an antecedent already mentioned in the

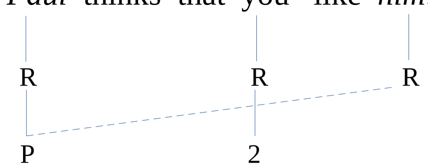
4 In fact, the *R*-node is the image of the thematic node, as discussed in Desouvrey (2024b). If an element does not have thematic morphology (also known as case morphology), it may not have its own *R*-node. This point seems to be relevant only in languages with declension morphology and will be ignored here.

discourse in order to be interpretable. Pronouns are thus free but not autonomous. The dashed line is used to indicate that the pronouns get the RFs *J* and *P* by spreading from an antecedent already mentioned in the discourse.

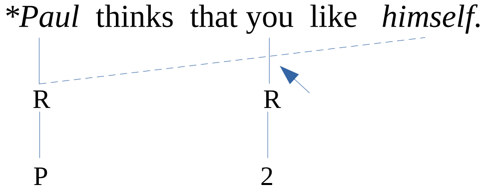
(3) a. Paul likes him. b. He likes Mary.

It is important to note that the *R*-nodes, which are grammatical features, are aligned on the same tier and thus make up a plane (see below). On the other hand, RFs are virtual features presumably pointing to the location of real world entities in the discourse space-time. Therefore, they do not pair up and may not align to make up a tier. This is illustrated with the following structures. In (4a) the RF *P* is assigned to the third person pronoun *R*-node by spreading. The spreading line does not cut through the association line of the RF [*2*], which presumably points to the real world individual. In (4b) on the other hand, since the *R*-nodes pair up and make up a plane, the spreading of the higher *R*-node violates the line-crossing ban, as indicated by the arrow pointing to the intersection.[5] If there is no intervening *R*-node between *Paul* and the anaphor, a perfect structure obtains, as shown in (4c).

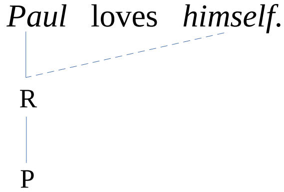
(4) a. *Paul* thinks that you like *him*.



b. **Paul* thinks that you like *himself*.



c. *Paul* loves *himself*.

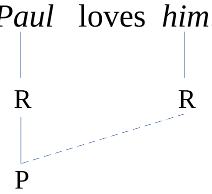


Now the matter is to account for the ill-formedness of the structures in (5a-c) in which *Paul* correctly assigns its RF to the pronoun. Unlike anaphors and NPs, third person pronouns are ambiguous as they can catch any RF in the discourse as long as it matches their gender and number. Based on this

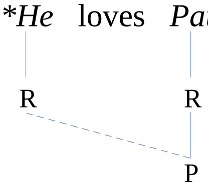
5 The ban on line crossing is a well-formedness condition on the nonlinear representation proposed in Goldsmith (1976). It might be that the spreading does not occur when an association line intervenes between a giver and a receiver, The line-crossing effect will be shown anyway, as in (4b), for the sake of clarity.

observation, Desouvrey (2003) has proposed a constraint against such ambiguity, which is stated below as (6). Thus, this constraint strongly excludes (5a) and (5b), where the pronoun supersedes an anaphor (cf. 4c). On the other hand, I assume that (5c) is weakly excluded because the use of a pronoun cannot be avoided with an equivalent structure, (5d). That is, if the ambiguity is removed, sentence (5c) will become acceptable (see below).

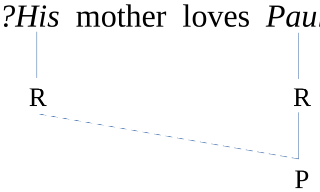
(5) a. **Paul* loves *him*.



b. **He* loves *Paul*.



c. *?*His* mother loves *Paul*.

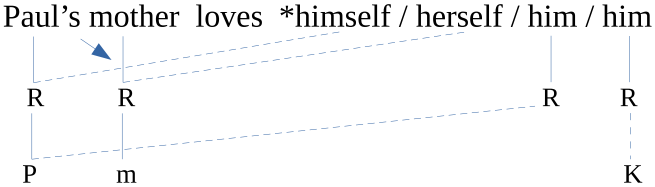


d. *Paul's* mother loves *him*.

(6) **Anti-ambiguity Constraint** (adapted from Desouvrey 2003)
Use of a pronoun must be blocked whenever an anaphor is possible.

The following structure further summarizes the property of each type of referring element as well as the effect of the nonlinear representation. In the possessive phrase, only the right-hand element can antecede an anaphor. On the other hand, *Paul* may or may not antecede the pronoun *him*, which is not constrained by the ban on line crossing.

(7) Paul's mother loves *himself / herself / him / him

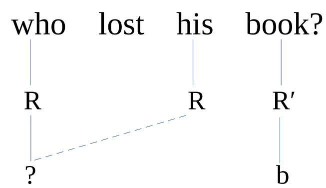


Summing up, under this view of coreference, an NP is absolutely free in any structure it appears in, as it carries its own RF. A third person pronoun is also free, but it must have a RF from an antecedent. Since there is no constraint on the location of the antecedent, which can be far away, a third person pronoun is inherently ambiguous. On the contrary first and second person pronouns have their own RF, which are represented with [1] and [2] respectively. Quite contrary, an anaphor can only have a local adjacent antecedent in order to avoid a line-crossing effect. Therefore, it is not prone to any ambiguity effect.

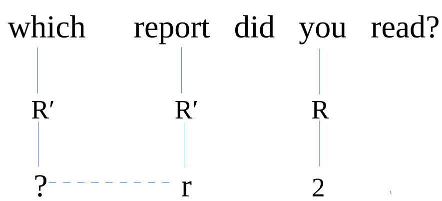
2.2.2 Functional features

Wh-operators are used to identify an unknown referring element. Thus it must be the case that they are specified for a special feature indicating that some element exists but it is unknown to the speaker. Such feature will be denoted by an interrogation mark [?] (formerly *x* in Desouvrey 2003). I will refer to this type of feature as a *functional feature* (FF), as opposed to RFs, which denote an existing real world entity. FFs must eventually pair up with the RF of the noun they modify. Thus, in (8a) the feature [?] of the operator is normally assigned to the pronoun as a RF. However, used as a determiner, it is a FF and it pairs up with the RF of the NP it modifies, as shown in (8b). Notice that I assume that inanimates (or perhaps nonhuman) have a distinct referential node, which is referred to as *R′*.

(8) a.

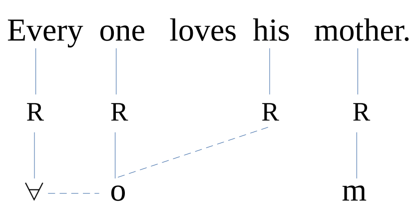


b.



Similarly, quantifiers must be differentiated from one another by their functional feature: ∀ (universal), ∃ (existential), ¬ (negation). These features modify the RF of an NP by pairing up with it, as shown below. Notice that contrary to the orthography, *everyone* (and similar words) must be represented as two separate words for the sake of clarity.

(9)



To conclude, the relation between two referring element is realized by spreading. The dependent element, or antecedee, must receive a referential feature or an *R*-node from its antecedent. Thus, it must be the case that the antecedent is already present in the structure or the discourse, meaning in the normal case the antecedent necessarily precedes the antecedee. The reverse case is certainly possible, but it is very marked, and in addition it presumably conveys some special stylistic effect (see below).

3. The vector theory

In Desouvrey (2000) it is observed that pronominal clitics in Spanish fail to move under a syntactic analog of the Obligatory Contour Principle (OCP) when they are complement of infinitive verbs.[6] This leads to the assumption that both infinitive verbs and clitics are operators in Spanish. Under this view, a conflict takes place between the OCP and the Superiority Condition: the former wants to move the clitics out of the VP domain, while the latter opposes this movement. In subsequent work, I suggested that *wh*-operators and negation are universally specified for a feature referred to as [ω], as opposed to other elements which are either neutral [∅] or specified for [φ] in their root nodes. The generalization that follows is that the feature [ω] may not be the privilege of operators, as it can be found in other categories on a language particular basis. Since ω-specified elements exhibit scope and other special properties irrespective of their categories, they are considered as vectors.

English lexical verbs, as well as modals and auxiliaries, are specified for the feature [ω], hence their difficulty with negation and *wh*-operators, which is overcome by the mediation of *do*. Accusative-specified operators must exit the VP under OCP, but their movement may not take place across the vector lexical verb. In Desouvrey (2007), it is suggested that *do* is used as a dummy vector in order to avoid a violation of the Superiority Condition (Chomsky 1973), which is here subsumed within the Barrier principle (Desouvrey 2024b), as stated in (10).

(10) **Barrier Principle**
A lower vector can neither move nor spread its feature across a higher vector in its domain.

Let us illustrate this vector effect with the derivation of (11a). Assuming that English is an SVO language, if this sentence were to be derived from an input like the one in (11b), an insuperable Barrier effect would take place, as the vector operator has to move across another vector, the tensed verb. Instead, an alternative input with a dummy vector is used as shown in (12a), while the lexical verb takes a neutral morphology, meaning it is no longer a vector.[7] From this structure, the operator is fronted (12b), and then finally the dummy vector *did* is adjoined to the operator. The resulting structure (12c) is well-formed because the *wh*-operator is no longer a vector, given the principle in (13), and as a result the Barrier effect induced by the movement across the dummy vector is canceled. Under this type of adjunction, conveniently represented by the sign '=' in the linearized structure (12d), the adjunct ends up on a different tier, t_2. (see e.g. Desouvrey 2020, 2021 for some details and consequences). In addition, the adjoined element and its host are welded together, making up a two-tier bimorphemic element. This fact, together with (13), contributes to eliminate the Barrier effect, since the out-scoped vector becomes a part of the fronted vector.

---

6 According to the syntactic OCP, elements specified for the case assigned by the verb must exit the VP domain in SVO languages. Thus, English adverbs and *wh*-operators, which are specified for accusative case, unlike personal pronouns, cannot stay in the postverbal position, just like clitics in Romance (see Desouvrey 2000, etc.). In SOV languages, object clitics presumably cannot adjoin to the subject, given the ban on vacuous movement.

7 I assume that the simple form of the verb is neutral, but the infinitive marker *to* is a vector. Thus, the negative morpheme can move across the simple form of the verb in its subjunctive use without the mediation of *do*:

(i) He insists that we not call Mary.

(11) a. Who did you see?

b. *[You [saw who?]] (failed input)

(12) a. [you [did [see who]]] (alternative input)

b. Who [you [did [see t]]] (Barrier effect)

c. *Who* [you [*t* [see *t*]]] _ _ _ _ t_1 (neutralization of the vectors)

did _ _ _ _ t_2

d. Who=did you see? (linearized structure)

(13) **Vector Neutralization Principle**
If a vector is adjoined to another vector, both become disabled,
i.e. they lose their vector properties.

However, in a structure with multiple vectors the Pendulum Principle stated in (14) (cf. Desouvrey 2021, 2022) comes into play.[8] Consider (15a), whose input is as in (15b). The *wh*-operator moves from the complement clause to the front of the matrix clause, and then the adjunction of *do* takes place. The movement of the operator across the embedded verb (*likes*) does not violate the Barrier principle, as it ends up in a different domain. Unlike the simple case discussed above, this structure contains three vectors (*who*, *do*, and *likes)*, which are not aligned on the same tier (cf.12c). In this situation, the lowest vector becomes the resultant of the higher ones. In addition, the lower vector alternately entangles with each vector in the pair, yielding a structure with two states, as shown in (15b,c). As a result, the operator and the modal remain active in both states.[9] As I will show, the crossover effect is sensitive to structures with multiple states.

(14) **Pendulum Principle**
Vectors alternately entangle with one another, resulting in a structure with multiple states.

(15) a. Who do you think that Mary likes?

b. [you do think that] [Mary likes who]

c. Who = do [you think that] [Mary likes *t*] (state 1)

8 The Pendulum Principle is not completely understood at this point. The structures in (15c,d) show the entanglement of the morphemes, but it is likely that every vector, whether it is a referring element or not, bears a special feature by which their entanglement takes place.

9 Notice that any further vector intervening between these three vectors are likely to be eliminated by a bypass effect induced by the Parallelogram Law, as discussed in Desouvrey (2021, 2022).

d. Who = do [you think that] [Mary likes] (state 2)

A further interesting property of vectors, which is relevant for the crossover phenomena, as will be shown, is the orientation of their spreading, which cannot change throughout the derivation. Let us refer to this property as the Directionality Constraint, as stated in (16) (see also Desouvrey 2008).

(16) **Vector Directionality Constraint**
Feature spreading from a vector is permanently oriented in only one direction throughout the derivation, either left-to-right or right-to-left.

With these assumptions in mind, let us turn now to the analysis of the crossover effect, which is based on data discussed in the literature.

4. Crossover effects

I will show that the crossover effects occur under the following conditions: (a) a *wh*-operator is used in a structure to extract an anaphor (SCO), and (b) its movement across its dependent induces a transitory Barrier effect and a Directionality effect (WCO). SCO effects with quantifiers are due to a line-crossing effect, while WCO is the result of the Pendulum effect by virtue of which the FF of the universal quantifier alternately entangles with every NP in its domain. On this view, only *wh*-elements and quantifiers are involved in the crossover phenomenon. In the generative framework, Lasnik and Stowell (1991) has made a similar claim, which is contested by Postal (1993). I consider Postal's data in a later section, showing that the observed distortion is caused by a further property of the nonlinear representation, namely a constraint requiring that both the antecedent and the antecedee be in the same plane.

4.1 *Wh*-crossover

The classical sentences (1e,f), repeated below as (17), illustrate a SCO and a WCO respectively. Traces and indices are conveniently used as descriptive devices, just like italics, and hence they do not have any theoretical relevance.

(17) a. *Who$_i$ does he$_i$ love t_i ? (SCO)
b. *?Who$_i$ does his$_i$ mother love t_i ? (WCO)

In this theory, a derivation starts from an input structure to which further transformations apply.[10] It must be the case that there are constraints on the formation of the input. I assume that the input for an interrogative sentence must arise from a transformation on a related basic sentence, which I will refer to as the *primitive*. On this view, the input for (17a) must be as in (18a), ignoring the failed input without *do*, as discussed above. The primitive of such an input can only be (18b), but neither (18c) nor (18d), which are not possible, as seen above ((18d) will be further discussed below). It turns out that

10 The input is built by merging lexical elements, which are fully inflected if necessary. This theory does not use functional heads, abstract levels of representation like LF, structural relations like c-command, and traces. What is important is the lexical elements and their features, which are organized in a tree, much like in nonlinear phonology.

this input (18a) is not faithful to its primitive. In effect, the *wh*-operator, which is an autonomous element with a *R*-node and a RF [?], comes to stand for a dependent element with a bare root node. As a result, the dependency relation is altered and, in fact, it is reversed in the input. I claim that this type of input is ruled out by the constraint in (19), which implies that an *R*-element, whether it bears a RF or not, cannot stand for an anaphor in any derivation. This principle accounts for the strong *wh*-crossover effect in simple sentences. A further type of strong crossover induced by vector interaction will be seen below.

(18) a. *[*he* [does [love *who*]]]
b. *he* loves ***himself***.
c. **he* loves *him*.
d. **he* loves *Paul*.

(19) **Dependency Inalterability Principle**
In any derivation, the relation of dependency in the primitive structure may not change throughout the derivation.

In addition, if the derivation still proceeds from this distorted input, further problems arise, as can be seen in (20). In the input (20a), the operator spreads its feature leftwards across the higher vector in violation of the Barrier Principle (cf.10). In the next derivational step, the operator is fronted, which again gives rise to a second Barrier effect, and a Directionality effect as the spreading is now reoriented rightwards. Finally in the output, all vectors are disabled (cf. 13). I claim that this structure is very bad because its derivation contains too many transitory constraint violations, three in all, and a major flaw in the input, which is not consistent with its primitive.

(20) a. [He [does [love who]]] (transitory Barrier effect)
R R
?

b. Who [he [does [love *t*]]] (transitory Directionality and Barrier effects)
R R
?

c. *Who*=does [*he* [*t* [love *t*]]] (output: neutralization effect)
R R
?

Consider now the derivation of the WCO example in (17b), as shown in (21). The primitive of this input (21a) may be a either an NP or a pronoun (the best alternative), as shown in (22). The anaphor is excluded by a line-crossing effect under the relevant interpretation, as discussed above. Both pronouns have the same antecedent, which must have been mentioned earlier in the discourse. Hence the *wh*-extraction of the lower pronoun *him* does not really trigger a violation of (19) to the extent that the pronoun (*his*) still depends on its original antecedent, which the *wh*-operator reinforces. The derivation normally proceeds from (21a), which contains a Barrier effect as the operator spreads its feature across the dummy vector. In the intermediate step (21b), a further Barrier effect takes place by the fronting of the operator, which results in a Directionality effect. The last step is faultless as the vectors are disabled (21c). Yet the sentence is not perfect. The transitory Barrier effect triggered by the movement of the operator is normal in English *wh*-structures and therefore can be ignored. Thus, it must be the case that the culprit is the transitory Directionality effect. This leads to the formulation of the WCO condition in (23). (SCO condition will be stated at a later point.)

(21) a. *His* mother does love *who* (Barrier effect)

R R

?

b. *Who* *his* mother does love (Barrier and Directionality effects)

R R

?

c. *Who*=does *his* mother love? (Neutralization effect: WCO)

R R

?

(22) *His* mother loves *Paul* /*him* /**himself*.

(23) **Weak Crossover Condition**

WCO arises if either (a) or (b):

a) The derivation includes multiple transitory constraint violation.

b) One state of the structure is ill-formed (see below).

This analysis is supported by the fact that if the *wh*-operator is not disabled by the adjunction of *do*, the crossover effect is strong despite the irrelevance of (19). This situation occurs in structures with a complement clause like (24a) (from Safir 2015:1). The input for this structure (24b) is faithful to its primitive, *He said Mary saw him*. However, the Barrier principle is violated, as the operator spreads its feature across two higher vectors. In the next step, the operator moves out of its domain to the front of the higher clause (24c), causing a further instance of the Barrier effect in its new acquired domain. In the final step (24d), the pair of operators is not disabled, as a pendulum effect takes place (cf. 14).

According to this principle, the third vector (*saw*) alternately entangles with each vector in the pair, preventing them from neutralizing each other. As a result, the *wh*-vector remains active in the structure, and thus the output contains both a Directionality effect and a Barrier effect, which accounts for the SCO judgment.

(24) a. **Who* did *he* say Mary saw? (SCO)

b. [*he* did say] [Mary saw *who*] (Barrier effect)

R R

?

c. who *he* did say Mary saw? (Barrier and Directionality effects)

R R

?

d. *who=did *he* say Mary saw? (output with active *wh*-vectors)

R R

?

Before concluding this section, consider the pairs of sentences in (25), which are adapted from Ishii (2006:158) (see also Agüero-Bautista 2012). I assume that a complementation structure consists of a matrix relative clause and a main clause, making up two islands (Desouvrey 2008 and earlier work). In both sentences, the *wh*-phrase moves from the complement clause to the front of the matrix clause. Ishii points out that WCO occurs only if the dependent pronoun is in the matrix clause (25b). If the pronoun is in the complement clause, the WCO effect disappears (25a). On the other hand, if the *wh* is not a determiner, WCO occurs irrespective of the location of the dependent pronoun, as shown in (26).

(25) a. [Which *student* do you think that] [*his* teacher scolded *t* in yesterday's geology class]?

b. *?[Which *student* does *his* teacher think that] [Mary scolded *t* in yesterday's geology class]?

(26) a. *?[*Who* do you think that] [*his* teacher scolded *t* in yesterday's geology class]?

b. *?[*Who* does *his* teacher think that] [Mary scolded *t* in yesterday's geology class]?

The sentences in (26) are bad because of the Directionality effect in the output, which is due to the activity of a third vector, the embedded verb, as discussed above. This can be seen in the derivation of (26a), for instance. It proceeds from input (27a), where the operator spreads its feature across the verb in violation of the Barrier Principle. Then, the operator is fronted, yielding a Directionality effect and a

Barrier effect (27b). In the output (27c), none of the vectors is disabled, and hence the directionality effect persists.

(27) a. [You do think that] [*his* teacher scolded *who* in yesterday's …]?

R R R R

2 t ?

b. [Who you do think that] [*his* teacher scolded *t* in yesterday's …]?

R R R R

? 2 t

c. *?[Who=do you think that] [*his* teacher scolded *t* in yesterday's …]?

R R R R

? 2 t

Let us turn now to the sentences in (25). I claim that a Pendulum effect takes place so that the FF of the *wh*-determiner alternately entangles with the RF of any dependent phrase in its domain (its clause). Let us suppose that the entanglement of the *wh*-determiner must conform with the condition in (28). With this in mind, consider the derivation of (25a), which proceeds from input (29a). The FF of the *wh*-determiner is permanently entangled with the RF [s] because the dependent RF [t] in this domain is on its opposite side and hence not under its scope. In addition, [s] cannot spread to the pronoun because it pairs up with the FF of the operator. In effect, the figure consisting of the nodes RR?s makes up a plane so that the spreading of [s] is blocked by the association line *?R*. Thus the dependency does not exist in the input.[11] In the next step, the *wh*-phrase exits its domain, triggering a Barrier effect in the matrix clause, which is canceled by the incorporation of *do* to the operator. It appears that the escape of the *wh*-operator in another domain where it does not have any dependent clears up the problems in the complement clause while allowing it to start life anew without any bad history. Therefore, in the resulting structure (29b), the *wh*-phrase comes to antecede the pronoun as though for the first time. There is thus no Directionality effect in this derivation, which results in a WCO-free sentence.[12]

(28) ***Wh*-determiner Entanglement Condition**

A *wh*-determiner normally entangles with a dependent NP if either (a) or (b):

---

11 Normally coreference must exist in the input and cannot be altered in the course of the derivation, as assumed in previous work. If WCO is absent in (25a), such a requirement may be too strong or else it depends on other contingency, like a relevant matrix clause (cf. fn. 12).

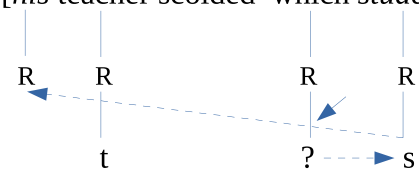
12 In the absence of a matrix clause, WCO is expected to show up. The following sentence (Postal 1993:549) is similar to the complement clause of (29). However, the *wh*-phrase does not have any other domain to escape, which results in a WCO.

(i) *Which *lawyer* did *his* clients hate? (WCO)

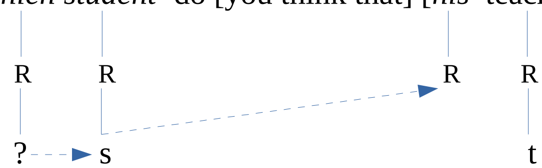
a) The NP is in a different domain.

b) The NP is in the domain of the *wh*-determiner and under its scope.

(29) a. [you do think that] [*his* teacher scolded which *student* in …] (input)

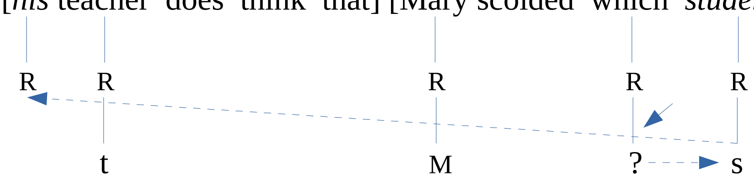


b. *which student*=do [you think that] [*his* teacher scolded *t* in …] (output)

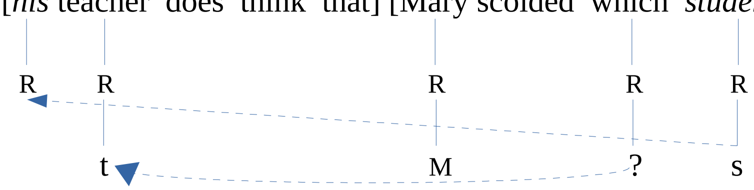


Consider now (25b). The FF of the operator is alternately entangling with both RFs, which are not in the same domain. This Pendulum effect yields a two-state structure. In one state (30a), the FF of the *wh*-operator is entangled with the RF [s] of *student*, which thus cannot antecede the pronoun, given the ban on line crossing. In the second state (30b), the FF is entangled with the RF [t], which is in a different domain, and hence *student* can spread its RF to the pronoun. Each state gives rise to a different derivation. A derivation from the first state, where the dependency fails in the input, is similar to that of (29) and hence is acceptable. On the other hand, a derivation from the second state yields a persistent Directionality effect. In the input, the pronoun is anteceded leftwards, as can be seen in (30b), whereas in the output the spreading is redirected rightwards, hence the WCO effect, as shown in (31).

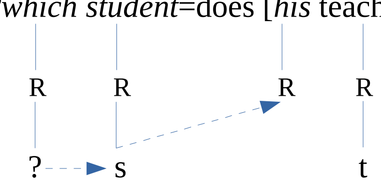
(30) a. [*his* teacher does think that] [Mary scolded which *student* in …] (state 1)



b. [*his* teacher does think that] [Mary scolded which *student* in …] (state 2)



(31) *?*which student*=does [*his* teacher think that] [Mary scolded *t* in …]

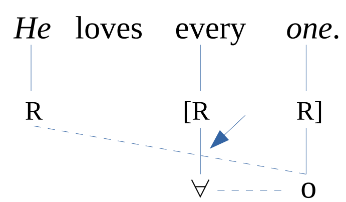


Summing up, with respect to *wh*-operators, SCO is due either to the alteration of the dependency in the primitive structure or to a Barrier effect and a Directionality effect induced by the activity of a third vector. On the other hand, WCO arises when there are transitory constraint violations in the derivation, including a violation of line crossing ban (see below).

4.2 Quantificational phrases and crossover

In the case of quantifiers, repeated in (32), the situation is different. As suggested above, the FF of the quantifier pairs up with the RF of the noun, and therefore they make up a plane. Thus, if a noun precedes the quantifier phrase anteceding it, a line crossing effect results in, as shown in (33). Clearly, the figure *RR*∀ *m* can be seen as making up a rectangle pivoting on the tier *R-RR*. Thus any line from [o] to the external *R*, which aligns itself with [*RR*], must intercept the *R*∀ side. Since this structure cannot be repaired, for instance by moving up the QP, a SCO obtains.

(32) a. **He* loves *everyone*. (SCO)
b. *?*His* mother loves *everyone*. (WCO)

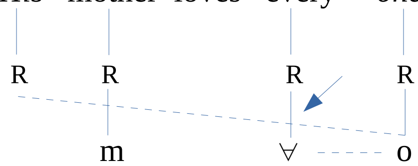
(33)


Consider now the WCO example (32b). This structure differs from the SCO one in that the dependent pronoun is a constituent of a phrase, just like *wh*-WCO. The universal quantifier morpheme is not a vector, but it appears that its functional features behaves like that of the *wh*-determiner in many respects, as I will show.[13] Let us assume that the FF of the universal quantifier entangles alternately with every RF in its domain.[14] Thus, (32b) must have two states, as shown in (34). In (34a), the quantifier is entangled with the RF [o] of its complement, which yields a line-crossing effect similar to (33), and hence this state is not acceptable. It should be recalled that the RF *m* is not in the way of the spreading line, as it is not aligned with the QP tier. In (34b), the quantifier pairs up with the RF [*m*], and both make up a plane. Since the feature [*o*] is not aligned with the QP tier '*m*∀', it can reach the pronoun without inducing a line-crossing effect. One can conclude that this derivation is not fully acceptable because one of the two states is a SCO. To put it another way, it is a fifty-fifty derivation.

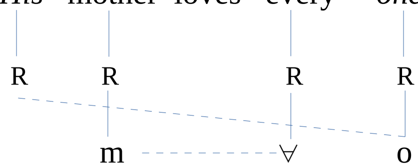
13 If the quantifier morpheme were a vector (ω-specified), it would not be able to antecede relative anaphor *that* in a matrix clause, just like *wh*-operators. The reason is that the φ-specified root node of *that* is incompatible with vectors. This can be seen in this example: [Paul thinks (*that*)] [*everyone* likes him], where the quantifier correctly antecedes *that* by spreading its *R*-node. Notice that the *R*-nodes being undifferentiated, I assume that when the terminal features pair up, the closest R-nodes spreads to the relative anaphor. There is thus no line-crossing effect.

14 On the other hand, the existential FF (∃) seems to be permanently entangled with the RF of the noun it modifies.

(34) a. *His* mother loves every *one*. (*state 1)



b. *His* mother loves every *one*. (√state 2)

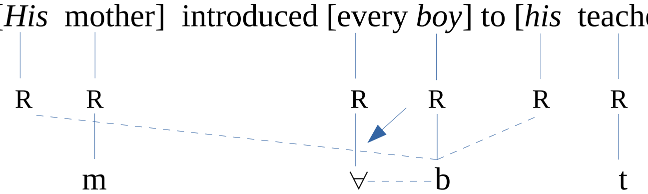


Under this analysis, the quantifier is not a vector, but its functional feature is subject to entanglement with every RFs in its domain. Further data support this analysis. Consider the following pairs of sentences, which are discussed by Agüero-Bautista (A-B)(2012:35) referring to Hornstein (1995). Sentence (35e) is similar to (34) and its acceptability is strongly distorted for the same reason: one state out of two is bad as the quantifier is oscillating between *boy* and *mother*, proper names like *Mary* being assumedly not quantifiable. In contrast, sentence (35f), which includes a further dependent and quantifiable NP, *his teacher*, is less deviant.

(35) a. *His$_i$ mother gave his$_i$ picture to every student$_i$.
b. His$_i$ mother gave every student$_i$ his$_i$ picture. (WCO)
c. *His$_i$ mother packed his$_i$ sandwiches for every boy$_i$.
d. His$_i$ mother packed every boy$_i$ his$_i$ sandwiches. (WCO)
e. *His$_i$ mother introduced every boy$_i$ to Mary.
f. His$_i$ mother introduced every boy$_i$ to his$_i$ teacher. (WCO)

I show that (35f) is not very bad because the QP antecedes two NPs, and thus the structure has three states, of which two are well-formed. In the first state, the quantifier is entangled with *boy*, which therefore can correctly antecede only one of the phrases, the lower one, as shown in (36a). In the second state, the quantifier is entangled with *mother* (*m*∀), allowing *boy* to correctly antecede both pronouns (36b). Finally in the third state (36c), the quantifier is entangled with teacher (∀t), and therefore both pronouns are correctly anteceded by *boy*. Since two states out of three are perfect, the sentence is less deviant.

(36) a. [*His* mother] introduced [every *boy*] to [*his* teacher] (*state 1)



b. [*His* mother] introduced [every *boy*] to [*his* teacher] (√state 2)

R R R R R R

m ∀ b t

c. [*His* mother] introduced [every *boy*] to [*his* teacher] (√state 3)

R R R R R R

m b

∀ t

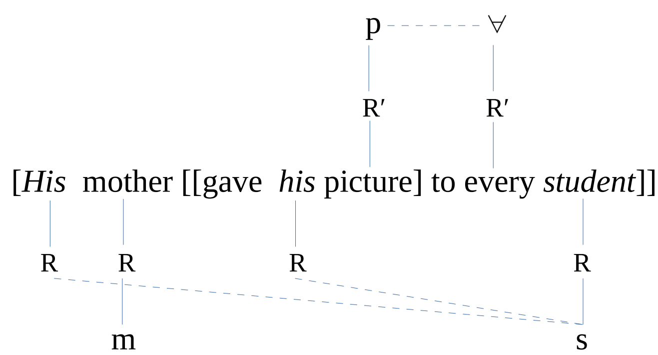
The two other pairs in (35) are similar. When both pronouns precede their quantified antecedent, a strong crossover obtains. The sentences are improved if the pronouns are on either side of the antecedent. These sentences present a further particularity: one of the dependent NP is not animate, unlike the pair (35e,f). As suggested above, inanimate phrases have their own referential articulation, *R′*. Being syntactic nodes, *R* and *R′* may neither pair up nor align themselves on the same tier. With this in mind, consider (35a), for instance. When the universal feature is entangled with [s], none of the pronouns can be anteceded, given the line-crossing effect, as shown in (37a). Hence this state is not acceptable. On the other hand, when the universal FF is entangled with [p], both pronouns are correctly assigned the RF of *student*, as shown in (37b). Notice that the referential articulation of the quantifier can be either *R* or *R′*, depending of its argument. Finally, in the third state (37c) all the *R*-nodes cannot align due to the intervening lone *R′*-node. If the lone *R′*-node could clear the way, the lone *R*-node would align itself with the *R*-plane (in brackets). Since this is not possible, it must be the case that the lone *R/R′*-nodes are in some peculiar state. I claim that any lone *R*- or *R′*-node is spinning around their root node, as indicated by the downwards arrow. In this condition, the *R*-nodes of *student* cannot align with the pronoun, which thus cannot get the RF of its antecedent. As a result, this state is ill-formed. This figure is different from (37a), where the lone *R′*-node is in between two pairs of *R*-nodes, hence two planes. It might be that the *R′*-node is either spinning, in which case there is an intermittent line-crossing effect, or forced out of the way by the planes. In any event, the ill-formedness of this state (37a) may not be due to the *R′*-node, as this type of configuration can be found in good sentences like (38), for instance. To conclude, this sentence is bad because two states out of three fail. The plane effect turns out to be an important condition on coreference; so let us state it in (39) (also see below).

(37) a. *[*His* mother [[gave *his* picture] to every *student*]] (*state 1)

[R R R] R′ [R R]

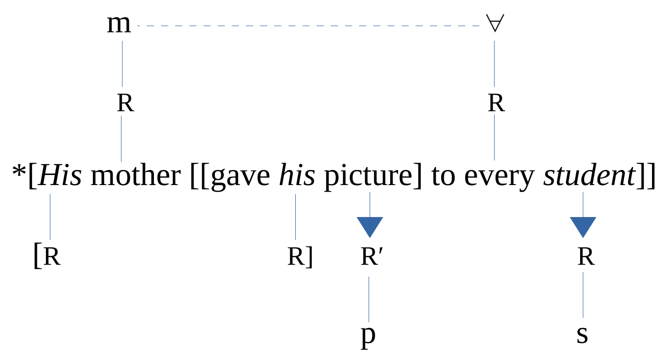
m p ∀ s

b.



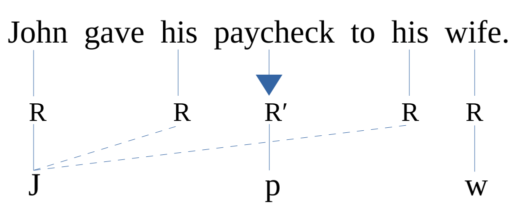
(√state 2)

c.



(*state 3)

(38)



(39) **Structural Conditions on Coreference**

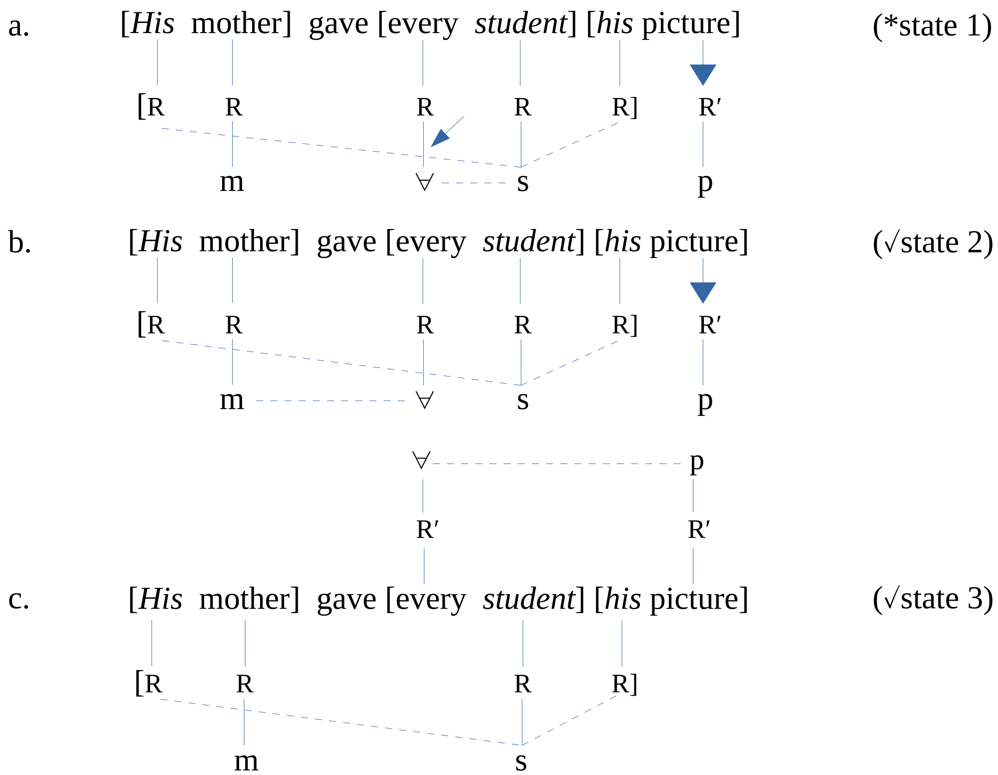
a) A pronoun and its antecedent must be in the same plane.

b) Two identical nodes can make up a plane if no other lone node intervenes between them.

c) A free node can align itself with a similar adjacent plane.

However, if the inanimate phrase is at the right edge of the structure, the sentence is improved as every *R*-nodes can make up a single tier, without any interference of the spinning *R′*. In (40a), where the quantifier is entangled with [s], the inanimate phrase can be anteceded, unlike the higher one, which is blocked by the association line of the quantifier feature. On the other hand, whether the quantifier is entangled with [m] or [p], as shown in (40b) and (40c) respectively, both phrases can be anteceded at the same time. Here again, two states out of three are well-formed, and therefore the structure is not very bad.

(40)

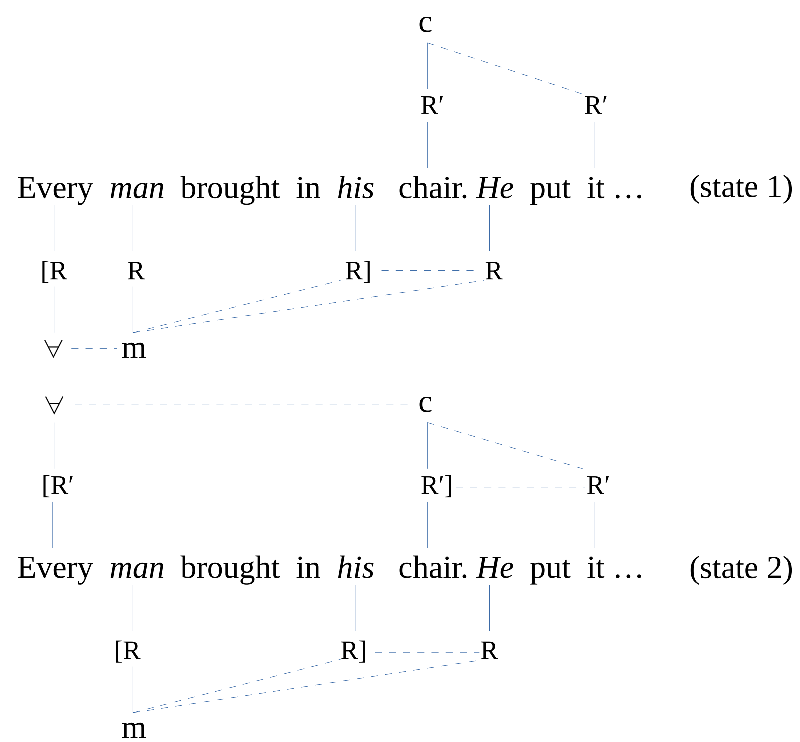


The paradigm in (35) is accounted for by the nature of the representation, as the degree of deviance of the sentences is due to a line-crossing effect and an intervening *R′*-node, which blocks the alignment of all the *R*-nodes. This analysis is supported by another phenomenon referred to as telescoping and discussed by A-B (2012, 2001) citing Poesio and Zucchi (1992). This is illustrated with the following sentences (his (74)). A standard assumption in the literature, which I assume, is that the field of action of a quantifier is limited to some domain, which is here the sentence. In the present perspective, (41a,c) are ill-formed because the *R*-node of the quantified NP cannot align with that of the pronoun it is intended to antecede, because it is in another inaccessible domain. Since the relation of antecedence is realized by spreading, the permanently quantified NP (*man*) cannot spread its RF outside of its domain.

(41) a. *Every *man* came in. *He* sat down.
b. Every *man* brought in *his* chair. *He* put it under the table.
c. *Every *man* sang. *He* whistled.
d. Every *man* sang a song. *He* whistled it.

On the other hand, in (41b,d) the quantifier is oscillating between two RFs in its domain so that none of them is permanently quantified. Consider (41b), for instance. In one state, the quantifier is entangled with *man*, which spreads its feature to *his*. At the same time, both *R′*-node are trying to align themselves so that pronoun *it* can be anteceded. This state of affairs forces the free *R*-node of the second sentence to align itself with the first sentence, instead of spinning, allowing the formation of the *R′*-tier, as shown in (42a). It appears that the formation of the *R′*-tier acts like a bridge unifying both domains, which allows the quantified noun to antecede *he*. Similarly, in the second state (42b), where the quantifier is entangled with *chair*, all *R*-nodes can be aligned on the same tier, and hence the nonquantified NP *man* can antecede both pronouns. Once all the *R*-nodes are aligned, making up a single plane, both sentences are unified. As a result, the lone *R′* of the pronoun aligns itself with the quantifier tier so that it can be anteceded as well. Since both pronouns are correctly anteceded in each state, due to the bridge effect, the sentence is acceptable.

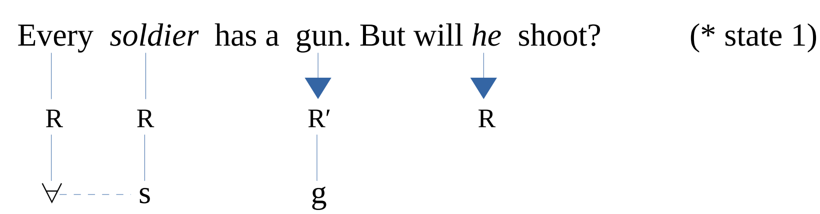
(42) a. / b.



This analysis correctly predicts that the bridge effect cannot take place if there is only one pronoun in the second domain. Consider the sentence in (43) (cf. Chomsky 1977, cited by Ruys 2000), which is not downright bad. I show that such a judgment obtains because in only one of the two states induced by the oscillating quantifier the pronoun can be anteceded. In effect, when the quantifier is entangled with *soldier*, the latter cannot reach the pronoun, which is out of its domain. This is shown in (44a), where the lone *R*- and *R′*-nodes are spinning in their respective sentence. On the other hand, when the quantifier is entangled with *gun* (44b), *soldier* is not quantified and hence can antecede the pronoun in the other domain. One state out of two is good, which explains the reported judgment.

(43) ??Every *soldier* has a gun. But will *he* shoot?

(44) a.



s

R R

b. Every *soldier* has a gun. But will *he* shoot? (√ state 2)

R′ R′

∀ g

Summing up, this analysis is based on the assumption that the universal quantifier does not permanently modify its complement. Rather the universal FF alternately entangles with every RF in its domain, yielding a structure with multiple states. The type of crossover depends on the number of well-formed states. The WCO effect appears when only one state is ill-formed, as stated above. As for SCO, it appears under the following condition:

(45) **Strong Crossover Condition**

SCO arises if:

a) The initial dependency is modified in the derivation.

b) The output contains a vector effect or a line-crossing effect.

c) At least two states are ill-formed in a multiple-state structure.

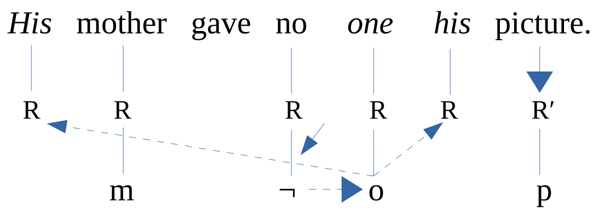
In addition, the interaction of the constraints on vectors as well as well as the properties of the representation, which groups animate and inanimate elements in different planes, makes it possible to vividly account for certain types of structures where a quantified NP must antecede two pronouns.

To complete this section, consider the case of the negative quantifier, as exemplified in (46) (A-B's (73)). A-B points out that the amelioration obtains in the pairs in (35) with the universal does not occur with negative quantifiers. Let us suppose that this quantifier, which inherits or perhaps mimics the properties of the negative morpheme (*not*)–universally a genuine vector, just like *wh*-operators–, must entangle with the RFs in its domain. However, unlike the universal, I assume that the entanglement of this vector-like quantifier is subject to the condition in (47). Consider (46a), for instance. When the quantifier is entangled with *one*, the higher pronoun is out of reach, given the prohibition of line crossing and the double directionality, as shown in (48a). When [p] is entangled with the quantifier, it correctly antecedes the inanimate NP, but not the higher NP, which is not under its scope (48b). Finally, the last state is not acceptable because the negative cannot entangle with the RF [m], which is out of its scope. The judgment indicates that (46b) is worst. This may be due to the fact that *one* belongs to a closed paradigm which includes non-vectors like *someone*, *everyone*, etc., while *boy* is just a dictionary entry. Therefore, by an harmonization effect (see below), speakers may be tempted to consider *one* as a neutral element, on a par with its counterpart in *everyone*, for instance.

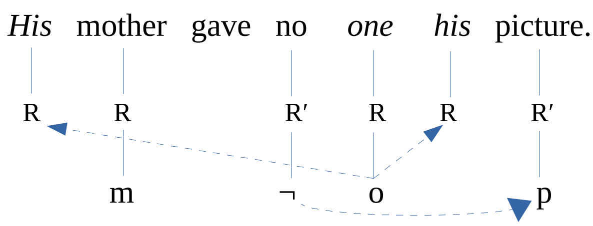
(46) a. *?*His* mother gave no *one his* picture.

b. **His* mother packed no *boy his* sandwiches.

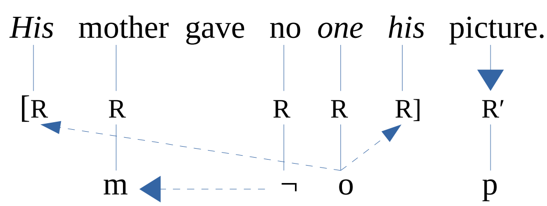
(47) **Negative Quantifier Entanglement Principle**
The negative quantifier must conform with both (a) and (b):
a) Its FF must entangle in turn with every quantifiable RF in its domain.
b) An entanglement is well-formed if the RF is under the scope of the FF.

(48)


(*state 1)

b.


(*state 2)

c.


(*state 3)

The following pair of sentences (from Postal 1993: 547) seems to support this analysis. According to Postal, the bracketed phrase in (49b), unlike the one in (49a), is a scope island. However, in the present perspective, it appears that in (49a) the quantifier is oscillating between *body* and *job*, which are both on its right side, hence the sentence is acceptable. On the other hand, in (49b) *opinion* and *body* are not on the same side of the quantifier, which results in an ill-formed sentence.

(49) a. I am sure [nobody's job] is important to Sidney.
b. *I am sure [your opinion of nobody] is important to Sidney.

4.3 The pair-list reading

A important body of research shows that quantifiers are ambiguous in certain contexts, as they allow in a question either an individual answer or a pair-list answer. This is illustrated with the following sentences (Safir 2015:26, referring to May 1985). The answer to (50a) may be either one individual or a list of pairs of individuals *x* and *y* such that *x* saw *y*. The pair-list answer is not available in (50b), which is thus not ambiguous. This phenomenon is considered as a type of crossover.

(50) a. Who did everyone see?
b. Who saw everyone?

Given our assumptions, one can observe that in (50a) the *wh*-operator is disabled by the adjunction of the modal (*who=did*), as seen above. Let us suppose that the quantifier feature can entangle with the feature of the disabled operator in a special way which amounts to fusion. That is, both features become complementary and the spreading of one is equivalent to the spreading of the other. Such an entanglement will be indicated by a left right arrow (↔). On this view, (50a) has two states: in one state the quantifier is entangled with the RF of *one* and in the other it is entangled with that of the operator, as shown in (51). I assume that the oscillation of the universal feature between [?] and [o] is what allows the pair-list reading (PL).

(51) a. Who=did every one see? (state 1)

R R R

? ∀ o

b. Who=did every one see? (state 2)

R R R

? ∀ o

To account for the case of (50b), one can assume that the feature of the live operator cannot interact with the FF of the universal quantifier; instead it is permanently aligned with the universal domain, which is indicated by the brackets in (52). For ease of exposition, let us say that in (52) the feature of the live operator *indexes* the quantifier domain. On this view the value of the operator is constant, just like that of the quantifier, which is permanently entangled with its complement. This state of affairs presumably blocks the pair-list reading. As a first approximation, one can conclude that the pair-list reading is a consequence of the successive entanglement of the quantifier with any number of compatible features, including the feature of the operator if it is disabled. If both the operator and the quantifier are permanently in one state, such a reading is not possible.

(52) Who saw every one? (*PL)

R R R

? [∀ o]

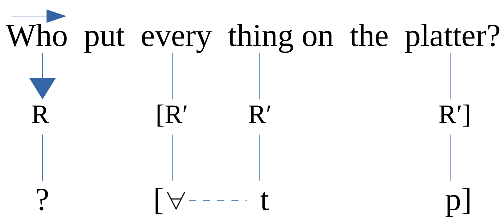
Consider now the following contrast (cf. Chierchia 1993:183-184), which shows a slightly different situation as the operators are active vectors in both sentences. (53a) allows the pair-list answer shown in (53b), whereas (54a) does not offer this type of answer (54b).

(53) a. Who put everything on the platter?
b. John, the chicken sandwich; Bill, the chow mein;…

(54) a. *Who* put everything on *his* platter?
b. *John, the chicken sandwich; Bill, the chow mein;...

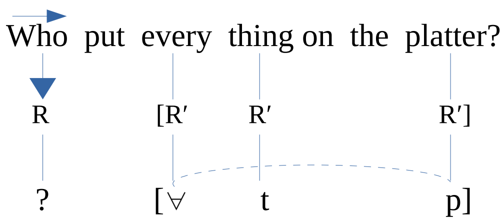
In (53a) the quantifier alternately entangles with [*t*] and [*p*], while the lone vector operator is spinning, as shown in (55). The spinning of the operator, I suppose, licenses the pair-list reading. In effect, if one assumes that a tier is an infinite straight line, as opposed to a segment, the spinning operator keeps intersecting the quantifier tier ∀-t/p. It appears that this intermittent contact amounts to indexing the quantifier. So, each time the operator intersects the quantifier tier, it presumably takes a different value, allowing the pair-list reading.

(55) a.



(state 1)

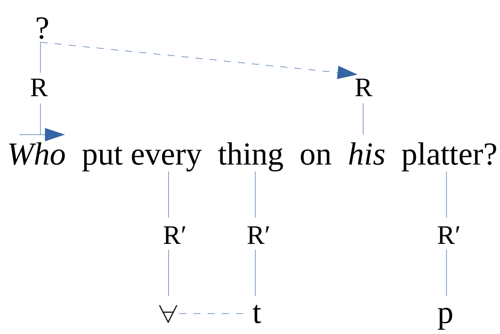
b.



(state 2)

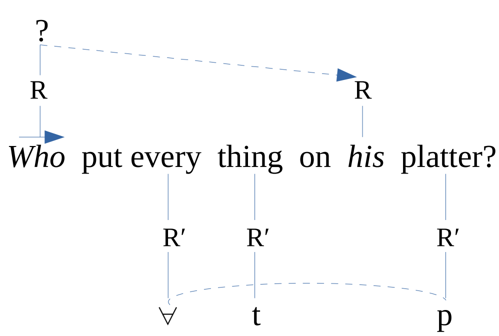
On the other hand, in (54a) the *R*-node of the operator makes up a different plane with that of the pronoun it antecedes, and hence the operator is not available to index the quantifier tier, as shown in (56). Thus even though the quantifier can entangle with *t* and *p* in its plane, the fact that its tier is not indexed by the operator blocks the pair-list answer.

(56) a.



(states 1)

b.



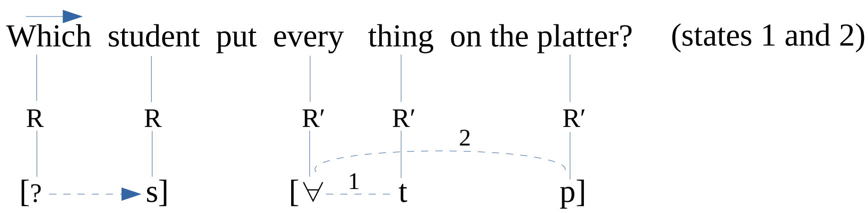
(state 2)

Now consider the following pair of sentences (Chierchia's (6) and (7)). As Chierchia points out, (57a) is similar to (53a) except that *who* is replaced by a *which*-phrase, which appears to be incompatible with the pair-list answer. However, a similar *which*-phrase in (57b) allows this type of answer.

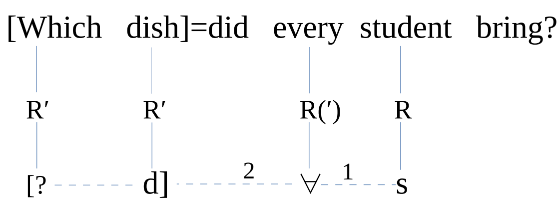
(57) a. Which student put everything on the platter? (*PL)
b. Which dish did every student bring t? (√PL)

Again, one can observe that in (57a) the *wh*-phrase is a live vector, while in (57b) it is disabled by the adjunction of the modal. In (57a), the live *wh*-phrase, which is in another plane, does not index the universal tier, as shown in (58). (Conveniently, two states are represented in the same structure, as indicated by the numerals.) As for (57b), the *wh*-phrase being not an active vector, it normally entangles in turn with the universal, yielding a second state, as shown in (59). Therefore, the pair-list obtains.

(58)



(59)



Under this analysis, the individual reading always exists, while the pair-list reading occurs under certain conditions, including multiple entanglement of the quantifier feature and its interaction with the *wh*-operator. This analysis seems to hold with the other universal quantifier, *each*, as well. Consider the following sentences, adapted from A-B (2012), who intends to show that the presence of *PRO*, in what is known as *PRO gate*, improves WCO. As indicated, the adjunct *PRO*-sentence easily allows the pair-list reading, unlike the adjunct possessive NP. I will show that what is crucial in this structure is not *PRO* as such, but the fact that it belongs to a sentential domain, unlike the possessive.

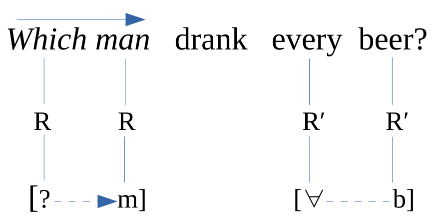
(60) a. *Who* convinced each judge by [*PRO* looking good]? (√PL)
b. *Who* convinced each judge by [*his* looking good]? (?*PL)

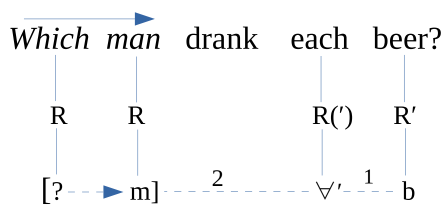
The universal quantifier *each*, I claim, is different from *every* in that it can entangle with live *wh*-vectors as well. Such a difference can be handled by assuming that *each* is the negative counterpart of *every*, given the assumption that vectors have poles (cf. Desouvrey 2022). Conveniently, the FF of *each* will be differentiated from that of *every* by a prime, ∀′. On this view, one can assume that *every* cannot entangle with live *wh*-features because both are positive (* +∀ vs +?). On the other hand, *each* with its negative pole can entangle with live *wh*-features (–∀′ vs +?). This property can be stated as follows:

(61) **The Functional Feature Repulsion Principle**
Two functional features with identical pole repulse each other.

The following minimal pair of sentences (cf. A-B 2012:36) strongly supports the Repulsion effect. In both sentences, the operator is an active vector; yet (62a) allows the pair-list reading, unlike (62b). In the latter, the universal cannot entangle with the live *wh*-phrase; it is rather permanently entangled with [b]. Thus both phrases make up their own plane, as they have different referential articulation, as shown in (63a). The quantifier does not interact with the *wh*-operator, and hence the pair-list reading is excluded. On the other hand, in (62a), the FF of the negative-pole quantifier alternately entangles with the RF of its complement and the *wh*-phrase, yielding a structure with two states, as shown in (63b). Therefore, the pair-list reading is possible.

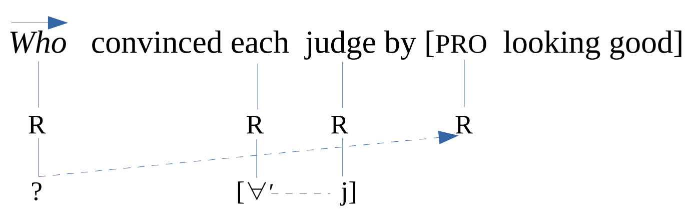
(62) a. Which man drank each beer? (PL)
b. Which man drank every beer? (*PL)

(63) a.


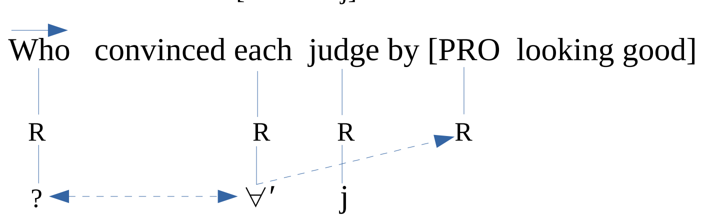
b.


(2 states)

Now, consider (60a). I assume that the interaction of a *wh*-vector with elements in different domains yields a Pendulum effect so that the vector is oscillating between both domains, each of which corresponds to a state of the structure. Thus, the *wh*-feature is either entangling with the quantifier feature or anteceding *PRO*, which is in a different domain, as shown in (64). Presumably each state allows the operator to take a different value, and hence the pair-list reading is possible. Notice that in (64b), the line-crossing effect is avoided by the fusion of the RFs of the operator and the FF of the quantifier, as suggested above.

(64) a.


(state 1)

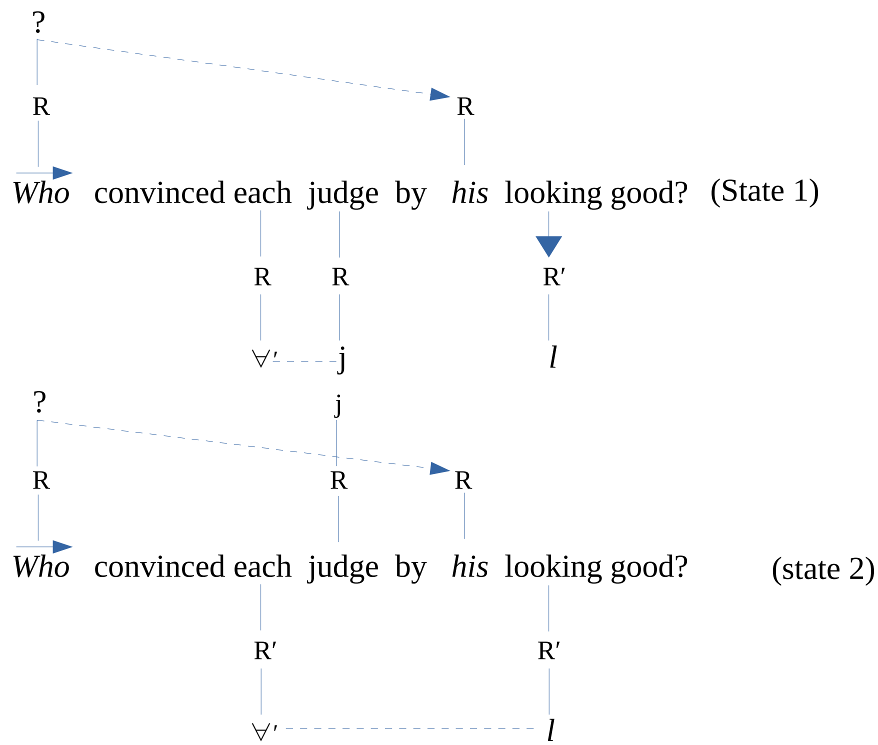
b.


(state 2)

In (60b), on the other hand, the *wh*-operator and the dependent pronoun are in the same domain, and hence the dependency relation is permanent, just as in (56) above. The universal does not have the scope to impose entanglement on the live operator. Whatever the state of the quantifier, the live operator is not available to entangle with it, as shown below. Therefore the pair-list reading is not possible.

(65) a.

b.



However, if the *wh*-operator is disabled, the pair-list reading obtains despite the dependency of the pronoun, as in (66) (adapted from A-B's (17a)). In this sentence the feature of the disabled operator enters in an entanglement relation with the universal feature, just like *one* and *adviser*, adding a third state to the structure. When the universal feature is entangled with either [o] or [a], the operator correctly antecedes the pronoun, as shown in (66b). Likewise, when the universal and the operator features are entangled, they enter in a complementary relation which amounts to fusion [?∀], and therefore the tier, as opposed to a single feature, spreads to the pronoun, as shown in (66c).

(66) a. *Who* did everyone introduce *t* to *his* adviser? (√PL)

?

R R

b. *Who*=did every one introduce to *his* adviser? (states 1 and 2)

R R R

2

∀ 1 o a

? ∀

R R R

c. *Who*=did every one introduce to *his* adviser? (state 3)

R R

o a

What appears to be crucial in the structure discussed above is the domain effect rather than the *PRO* effect. This can be further seen in the sentence in (67a), adapted from A-B's (17b). The *wh*-operator is extracted from the complement clause, which is in fact the main clause, and it is not disabled by the adjunction of the modal because the verb in the complement clause remains an active vector. Therefore, the *wh*-operator comes to deal with two domains, its original domain where it must antecede the pronoun and the larger domain where it takes scope over the quantifier. In one state (67b), the operator indexes the quantifier domain, while in the other it antecedes the overt pronoun (67c).[15] As for the universal FF, it alternately entangles with the RFs in both islands, which are unified by the dependency of the matrix relative via the deleted relative *that*.[16] Therefore the pair-list reading obtains easily.

(67) a. *Who* did every witness say *t* put the sandwiches on *his* plate? (√PL)

b. *Who*=did [every witness say ~~*that*~~] [*t* put the sandwiches on *his* plate?] (state 1)

R R R R′ R R′

? [∀ w s p]

15 It appears that the operator does not antecede the pronoun in all of the states. It seems that this situation does not affect the well-formedness of the structure.

16 On this view, there is no equivalence between the quantifier domain and the operator domain. For the quantifier, the relativization process unifies both clauses, whereas for the operator both clauses remain independent from each other.

?

R R

c. *Who*=did [every witness say ~~*that*~~] [*t* put the sandwiches on *his* plate?] (state 2)

R R R′ R′

[∀ w s p]

According to A-B, the pair-list answer is also available with quantifier *each* instead of *every*. A structure with this quantifier has several states, the three states of the operator plus the three states of the quantifier when the operator indexes its domain. In one state, the operator antecedes the pronoun, just like (67c). In the second state, it is entangled with the universal, since their poles are compatible (68a). In the third state, it indexes any entanglement of the universal with either [w], [s], or [p], as shown in (68b).

(68) a. *Who*=did [each witness say ~~*that*~~] [*t* put the sandwiches on *his* plate?]

R R R R′ R R′

? ∀ w s p

b. *Who*=did [each witness say ~~*that*~~] [*t* put the sandwiches on *his* plate?]

R R R R′ R R′

2

? [∀ 1 w s p]

3

Finally, if the above analysis is correct, the licensing condition of the pair-list reading can be stated as follows.

(69) **The Pair-List Reading Condition**

The pair-list reading occurs if:

a) The operator and the universal are not in a permanent state.

b) The operator and the universal interact by either indexing or entanglement.

5. More on weak crossover

In this section, I consider the absence of the crossover effect in certain constructions frequently discussed in the literature, namely relative clauses, parasitic gaps, *tough*-movement, and topicalization.

5.1 Relative clauses

There is a controversy in the literature regarding the presence vs. absence of weak crossover in English relative clauses, as illustrated in (70) and (71), adapted from Lasnik and Stowell (1991: 698, 706), and

Safir (2015: 14), respectively. For instance (70a), unlike (71a), is not uniformly judged by all speakers, hence the sign %. However, it is uncontroversial that restrictive relative clauses with a quantified head show the effect, as in (70b,c) and (71b), as opposed to (71c). On the other hand, everyone seems to agree that there is no weak crossover effects in appositive relative clauses (70d).

(70) a. % A *man* [*who* [*his* mother loves *t*]] lives at home.
b. *Every *man* [that *his* mother rejected t_i]…
c. *No *house* [that [*its* owner sold *t*]]…
d. *John*, [*who* [*his* mother loves *t*]], lives at home.

(71) a. A *man* [*who* [loves *his* mother]] lives at home.
b. ?*A*ny man* [*who* [*his* mother saw *t*]]…
c. A*ny man* [*who* [*t* saw *his* mother]]…

It is likely that this controversy is due to dialectal variation. If so, in the present perspective, one can suppose that, for certain speakers, relative pronouns are vectors whereas for others they are not. Under our assumptions, English verbs are vectors, and hence *wh*-operators and negation have difficulty exiting the VP. Thus, the dummy vector *do* is used as a mediator. On this view, the question arises as to how the relative pronoun is extracted from its argumental position to the front of its clause in the vector dialect.

Consider first the analysis of relative clauses proposed in Desouvrey (2008) and earlier work. The relative clause and the main clause are generated as two independent structures, or islands, as shown in (72a). The relative morpheme has a floating, or recessive, *R*-node, and hence it can be used either as a pronoun or an anaphor. In the input, it is a pronoun having a long-distance antecedent, *the man*. Then it moves to the edge of its clause as a first step to be closer to its antecedent. Finally, the relative clause is adjoined to the antecedent as an anaphor, and then the structure is linearized, as shown in (72c). In this derivation, the relative pronoun is not a vector, and so it moves across its dependent pronoun and the verb without inducing a Barrier effect.

(72) a. [*A man* lives at home] [*his* mother loves *who*]
b. [*A man* lives at home] [*who* [*his* mother loves *t*]]
c. [*A man* = *who* [*his* mother loves *t*] lives at home]

Appositive relative clauses are derived in the same way, except that the adjunction host is assumed to be an empty timing unit, or skeletal *x*-slot, as shown in the following derivation.

(73) a. [*John* lives at home] [*his* mother loves *who*]
b. [*John* lives at home] [*who* [*his* mother loves *t*]]
c. *John*, *x*=*who* [*his* mother loves *t*], lives at home.

However, if the relative pronoun (RP) is a vector, it may not move across the verb, given the Barrier Principle, as seen above. Thus, just as in interrogative sentences, the original input fails. In such a case, I claim that the relative clause is base-generated in front of its clause while its argumental position is occupied by a null co-referent pronoun (∅). Let us illustrate this with the derivation of (74a) (from Safir 2015). From the input (74b), the relative clause is adjoined to the head, yielding (74c).

(74) a. the man who we saw…
[*the man*…] [we saw *who*] (failed input)
b. [*the man*…] [*who* [we saw ∅]] (alternative input)
c. *the man* = *who* [we saw ∅] …

It should be noted that the mediation of *do* is not possible in relative clauses because it would yield an overloaded derivation. In this theory, a well-formed derivation may not contain more than three steps including the input, according to the Derivation Equivalence Number (e.g. Desouvrey 2018, 2019 and earlier work). Thus, a derivation with *do* would yield four steps, as shown in (75).[17]

(75) a. *his* mother does love *who* (input)
b. *who his* mother does love (fronting of *who*)
c. *who*=does *his* mother love (adjunction of *do*)
d. *the man = *who* = does *his* mother love … (adjunction of the clause to the head noun)

The same analysis applies to appositive relative clauses. The derivation of (70d) proceeds from input (76a). Then the relative clause is adjoined to the *x*-slot, yielding (76b) after linearization.

(76) a. [*John* lives at home] [who [*his* mother loves ∅]]
b. *John*, *x* = *who* [*his* mother loves ∅], lives at home.

Let us turn now to the crossover case in (70), repeated as (77a). This structure includes what I will refer to as a *referential string*, i.e., a referring element (*man*) with two dependents (*his* and *who*), all in the same domain since the RP is incorporated to the head noun. I claim that the crossover effect is due to the violation of the constraint in (78). In effect, the output is not conform to the state of the original input, (77b). Indeed, the initial order of the referential string (*his*-*who*) is changed to *who*-*his*, hence the WCO effect. Notice that that (78) does not apply to (76c) because the head of the relative clause is not in the same domain as its dependents.

(77) a. % A *man* [*who* [*his* mother loves *t*]] lives at home.
b. [*A man* lives at home][*his* mother loves *who*] (failed input)
c. [*A man* lives at home] [who [*his* mother loves ∅]] (alternative input)
d. *A man* = *who* [*his* mother loves ∅] lives at home.

(78) **Referential String Invariability**
Given a referential string making up a domain, the relative position of its elements must be conform with the original input if either (a) or (b):

17 However, negation does not overload the derivation. In effect, the structure is processed from left to right, so that the negative morpheme and the relative pronoun can move in turn in the same derivational step (ib):

(i) a. *his* mother does love not *who* (input)
b. who_2 *his* mother does = not_1 love t_1 t_2 (negation movement and then *wh*-movement)
c. the *man*=*who his* mother does=not love … (adjunction to the head noun)

a) The referential string includes a vector as a co-dependent.
b) The head of the referential string is a quantified NP.

The same pattern appears in the derivation of relative clauses with a quantified head like (79a) (=70b). I assume that the incorporation of the relative pronoun into the head noun allows the quantifier to take scope over the relative clause, hence incorporating it in its domain. Yet the sentence shows a WCO because the referential string is not consistent with the original input (79b), in violation of (78).

(79) a. ?*[*Any man* [*who* [*his* mother saw t]]] …
b. [*Any man*… [*his* mother saw *who*] (failed input)
c. [*Any man*…] who [*his* mother saw ∅]] (alternative input)
d. [*Any man* = *who* [*his* mother saw *t*]…

Interestingly, it seems that there is no dialectal variation with relative *that*, which is not a vector. In fact, it is incompatible with vectors as it has a φ-specified root node, which explains why it is absent in indirect questions. The relative clause in (70b), repeated as (80a), is derived from the input structure (80b), where the relative pronoun moves easily to the front of its clause, as shown in (80c). For all speakers, the resulting structure (80d) is not consistent with (78), hence the WCO effect.

(80) a. *Every *man* [that *his* mother rejected t_i]…
b. [*Every man…*] [*his* mother rejected *that*] (input)
c. [*Every man…*] *that* [*his* mother rejected *t*]
d. *Every man* = *that* [*his* mother rejected *t*]… (WCO)

To conclude, it appears that the controversy about WCO in relative clauses arises because for certain speakers the relative pronouns, those which are syncretic with *wh*-operators, are treated as vectors. The WCO effect is due to the violation of the constraint in (78). This constraint also accounts for the WCO effect in relative clauses with a quantified head, which affect all speakers.

5.2 Parasitic gap

Parasitic gap structures (*pg*) are frequently discussed in the literature as a case where weak crossover is unexpectedly absent (cf. Lasnik and Stowell 1991, etc.). The first question to deal with is under which conditions a *pg* is possible. Descriptively, it appears that a *pg* arises in an adjunct clause as a missing object argument referring to an object operator in the main clause. Presumably, a subject *pg* may not be possible because such a position is normally occupied by *PRO*. This can be seen in the following contrast (Lasnik and Stowell 1991: 692-693), where *e* stands for the *pg*. In (81) the object operator in the main clause licenses the object *pg*, unlike the subject operator in (82).

(81) a. Who_i did Mary gossip about t_i [despite your having vouched for e_i]
b. Which man_i did you look at t_i [after Mary had spoken to e_i]

(82) a. *Who_i t_i gossiped about you [despite your having vouched for e_i]
b. *Which man_i t_i looked at you [after Mary had spoken to e_i]

Now the task at hand is to account for the difference between the two pairs of sentences. I claim that the operator is generated in both the main clause and the adjunct clause, so that the input structures for (81a) and (82a), for instance, are as shown in (83a) and (84a), respectively. The validity of these structures is supported by the fact that a pronoun may be used in their non-interrogative counterparts, as shown in (83b) and (84b). Since the *wh*-operators are identical, as they are intended to identify the same individual, I assume that the lower one is barred at the output while its features remain active.

(83) a. Mary did gossip about *who* [despite your having vouched for ~~*who*~~]
b. Mary gossiped about *Jane* despite your having vouched for *her*.

(84) a. *Who* gossiped about you [despite your having vouched for ~~*who*~~]
b. *John* gossiped about you despite your having vouched for *him*.

Thus, from input (83a) the operator moves to the left edge of the structure in violation of the Barrier principle induced by its crossing of *did*. Then in the final step the structure is corrected by the adjunction of the modal to the operator, which results in the neutralization of both vectors. Since the functional features from both instances of *who* are identical, they must pair up and fuse, yielding a well-formed sentence, as shown in (85) (the irrelevant *R*-nodes are conveniently omitted).

(85) Who=did Mary gossip about [despite your having vouched for ~~who~~]

? ?

As for input (84a), no movement needs to take place, as the operator is already in the highest position in the structure, as shown in (86). This structure crucially differs from (85) in that the subject operator is not disabled. Therefore, their features cannot pair up and fuse, and in fact, they repulse each other, given (61). This repulsion effect is indicated by the double arrow in (86). Therefore the *pg* structure is not well-formed.

(86) *Who gossiped about you [despite your having vouched for ~~who~~]

? ?

To conclude, it appears that the *pg* phenomenon results from the deletion of a lower adjunct vector under referential identity with the higher one. Still, their functional feature must pair up. This can be realized only if the higher operator is not an active vector, as is the case of *wh*-object argument. The absence of crossover in such structures is normal under the present theory.

5.3 *Tough*-movement

Lasnik and Stowell (1991) also discussed *tough*-movement as another type of construction where WCO is absent. The particularity of this construction is that the object of the verb can either stay in situ, in which case the sentence takes an expletive subject, or appear in the subject position, as illustrated in (87) (from Lasnik and Uriagereka 1988:146). The question is whether both sentences in (87) are derived from the same input. I claim that the expletive subject may or may not be present in the input. On this view, the expletive in (87b) is base-generated in its position. On the other hand, (87a) is derived from an input with a subject gap as shown in (88a). Then the complement of the verb is fronted to fill in the gap, yielding (88b).

(87) a. John is tough to please.
b. *It* is tough to please John.

(88) a. *e* is tough to please John.
b. John is tough to please *t*.

Now consider the case where the argument of the verb is a *wh*-operator. If the operator is generated in the argument position of the verb, a Barrier effect will take place, for it has to move across two vectors in the same domain, the verb and the preposition (*to*), as shown in (89). However, this structure cannot be repaired by adjoining the verb to the operator under a principle that bans vacuous movement in any derivation. In effect, the adjunction of *is* to the operator would not modify the input, and would yield the same linear sequence *who=is*, unlike interrogatives with *do*. Thus, I am led to suggest that the *wh*-operator in this construction is extracted via relativization with *that*. That is, the operator is base-generated in its surface position as the head of a *that*-clause, as shown in (90). The relative is then deleted because, by its root feature φ, it is incompatible with vectors (cf. Desouvrey 2024a,b).[18]

(89) **Who* is tough to please *t?* (Barrier effect)

(90) *Who* [is tough to please ~~*that*~~]
R R
?

Let consider now a more complicated sentence with a potential WCO effect, as in (91) (Lasnik and Stowell's 20). As seen above, the operator is base-generated in its surface position as the head of the relative clause, as shown in (92). There is no movement of a lower operator across its dependent, and hence there is no crossover effect induced by the violation of the Barrier principle.

(91) *Who* will be easy for us [to get [*his* mother] to talk to *e*]

18 *Wh*-extraction via relativization is obvious in Haitian, as the relative pronoun is not deleted (Desouvrey 2009). Compare the following pairs (the object relative pronoun is often omitted). This phenomenon also exists in some non-standard dialect of French: *Qui qui est venu?* (who who is come, 'Who came?).

(i) a. Kimoun ki telefone Marie? vs. *Kimoun telefone Marie?
Who who phoned Marie (roughly 'Who phoned Marie?)
b. Kimoun (ke) Marie telefone? vs. Marie telefone kimoun?
Who (that) Marie phoned (roughly 'Who did Marie phone?')

(92) *Who* will be easy for us to get *his* mother to talk to ~~*that*~~ ?

R R R

?

It turns out that the absence of WCO in *tough*-construction is normal since there is no movement across any vector.

5.4 Topicalization

In the present theory, WCO effects are caused by either the special syntax of vectors or the properties of the representation, for instance the ban on line crossing. As discussed in the literature, WCO effects also apparently arise from movement of neutral elements, as in topicalization. This is illustrated in (93) (cf. Lasnik and Uriagereka 1988:154, citing Chomsky 1977). In (93a) the NP moves easily as it has no dependent pronoun. In (93b), however, the NP moves across its dependent, which results in a strong crossover effect according to the authors.

(93) a. John_1, Mary_2 thinks I like e_1.
b. *John_1, he_1 thinks I like e_1.

Sentence (93b) is derived from the sentence in (94a), which is a distorted version of (94b). Now it is a matter to explain why this type of sentence with a proper name cataphora (see Ross et al. 2023), is not acceptable. I suggest that cataphora with proper names are normally ruled out by two principles. Firstly, the Harmonization Principle (95), which aims to minimize exceptions in the grammar, hence facilitating the language acquisition. The paradigm here is that of elements with a RF, which includes *wh*-operators, quantified NPs, and perhaps any DP (definite or indefinite). Since crossover effects appear when such elements follow any dependent, as discussed above, speakers should feel uneasy with every referring elements anteceding backwards. Secondly, as seen above, a blocking effect triggered by the constraint against ambiguity eliminates such sentences like (94c), as the grammar has at its disposal appropriate material to build a better one (94d).

(94) a. *He_i thinks I like John_i.
b. John_i thinks I like him_i.
c. *He_i likes John_i.
d. *John* likes *himself*.

(95) **Harmonization Principle (HP)**
A constraint applying to a number of elements in a given paradigm tends to be generalized to every element in that paradigm as long as no violation of other constraints results in.

However, HP is not a strong constraint. In effect, its violation may not yield a strong ill-formedness, as the harmonized element does not violate any grammatical principle.[19] In other terms, proper names show both types of crossover effects by sin of association with offenders. Thus, cataphora with proper names can be used for stylistic effects as long as the ambiguity is removed, somehow or other, as in (96a) (Ross et al. 2023: 544), which is as exceptional as referring to oneself in the third person. In fact, this sentence could be uttered by *Sue* herself. As can be seen in (96b), the neutral RF normally spreads backwards to the dependent pronoun.

(96) a. Sue is not a popular girl, but her$_i$ mother loves Sue$_i$ unconditionally.

b. … but her mother loves Sue…

R R

S

Consider now the following pair of sentences (cf. Postal 1993). I take the cataphora in (97b) to be acceptable thanks to the emphatic character of this type of matrix clause.[20] However, what makes the topicalization more or less acceptable is the property of the nonlinear representation, as I will show. As seen above, the dependency relation is possible if both the antecedent and the antecedee are in the same plane. Otherwise, a crossover-like sentence obtains by a side effect, as the pronoun is trying to take any available referential element as antecedent. The structure of these sentences are shown in (98). In (98a), the NP and the dependent pronoun are adjacent on the *R*-tier and thus make up a plane within which the spreading takes place, while the *R′*-node is spinning. Therefore, the sentence is acceptable. In (98b) however, the lone *R′*-node is spinning between the two *R*-nodes and hence blocks them from making up a plane. Therefore, the pronoun cannot normally get the RF of its intended antecedent (cf. 31), which results in a deviant sentence.[21]

(97) a. *Harry$_i$, a picture of him$_i$ fell on t_i.
b. (I am quite sure that) a picture of him$_i$ fell on Harry$_i$.

---

19 As noted in Desouvrey (2018), the Harmonization Principle is usually opposed more or less successfully by language regulators and highly literate people, which gives rise to archaisms in formal dialects.

20 Similarly, I assume that such words like *only* and *even* improve sentences with crossover effect by their emphatic character, as can be seen in the following contrast (cf. Postal 1993:549). Recall that the derivation of a sentence like (ia) is fine, but it shows a WCO effect because its derivation contains two transitory constraint violations, a Barrier effect and line-crossing effect. Thus it appears that such emphatic words like *even* redeems the sentence by overruling these effects.

(i) a. *Which *lawyer* did *his* clients hate t_i?
b. Which *lawyer* did even *his* clients hate t_i?

21 If the indefinite determiner bears the existential FF feature (∃), one has to assume that the whole DP is spinning. More generally, it might be that any QP (including DPs) is spinning if it has no dependent. Alternatively, one can assume that the indefinite determiner does not have a referential node of its own, and hence its floating FF fuses with the RF of the noun (∃p). In such a case the determiner would be like an anaphor getting its *R*-node from the noun. Further research is needed as well as the help of a native speaker.

(98) a. … a picture of *him* fell on *Harry*.

R′ [R R]

p H

b. **Harry*, a picture of *him* fell on.

R R′ R

H p

The same condition appears in the sentences in (99), where the dependent pronoun is embedded in a larger phrase. Sentence (99a) (absent from Postal 1993) is presumably not too bad. In effect, the *R*-nodes are arranged in two planes, one of which contains the pronoun and its antecedent, while the lone *R′*-node may be spinning or not (cf. the discussion of (37)), as shown in (100a). On the other hand, in the topicalized structure (100b) the pronoun and its antecedent can never make up a plane; whatever way the three adjacent *R*-nodes are arranged, the *R′*-node will be spinning between them and that of the pronoun. Indeed, while R_1 and R_2 (i.e. the *R*-node with RF *1* and *2*, etc.) can pair up to constitute a plane, R_S and *R* (of *him*) are separated by the spinning *R′*. If instead R_s is aligned with the R_1-R_2 plane, *R* is prevented from aligning itself with them by the lone *R′*-node.

(99) a. I am sure [your opinion of him_i] is important to $Sidney_i$. (presumably acceptable)
b. *$Sidney_i$, I am sure [your opinion of him_i] is important to.

(100) a. I am sure your opinion of *him* is important to *Sidney*.

[R R] R′ [R R]

1 2 o S

b. *Sidney, I am sure your opinion of *him* is important to.

R R R R′ R

S 1 2 o

To conclude, this type of crossover seems to be due to the violation of (39), as the pronoun and its antecedent cannot always be in the same plane. This effect of the plane may not give rise to a strong judgment because, by a side effect, the pronoun will try to catch any adjacent RF (see Desouvrey 2024a).

6. Conclusion

The account of the crossover phenomenon presented above is mainly based on the Dependency Inalterability Principle (SCO) and the Vector Directionality Principle (strong and weak *wh*-crossover). A further constraint, the Barrier Principle, partly accounts for weak *wh*-crossover. Crossover effects with quantifiers are due either to the number of transitory constraint violation in the derivation or the number of faulty states in a multiple-state structure induced by the oscillation of the universal feature between the RFs in its domain, as well as the prohibition of line crossing in the nonlinear representation. The same analysis, which is based on the universal feature entanglement, makes it possible to explain the presence or the absence of the pair-list answer. Beyond these two types of constructions, a distortion appears in topicalization structures; but it is in fact a consequence of the representation, which requires a pronoun and its antecedent to be in the same plane. More generally, all referential element must normally be higher in the structure than their dependent, given the Harmonization Principle, which aims to eliminate exceptions in any paradigm.

This paper is based on data discussed in the various versions of the generative framework (cf. Chomsky, 1981, 1995), which focus on the universal quantifier. It is not clear if other quantifiers are subject to the Pendulum effect. Also, the determiners, definite and indefinite, have to be investigated in relation to their features. It might be that they bear an existential FF which fuses with the RF of the noun they modify. This is left for future research. Finally one may note that the crossover effect, though interesting, is a laboratory phenomenon, as it is unlikely that it may be found in any natural corpus, as other available constructions make it possible to avoid backward spreading to a dependent.

References

Agüero-Bautista, Calixto (2001). Cyclicity and the scope of *wh*-phrases. Doctoral dissertation, MIT, Cambridge, MA.

Agüero-Bautista, Calixto (2012). Team Weak Crossover. *Linguistic Inquiry* 43: 1–41.

Chierchia, Gennaro (1993). Questions with quantifiers. *Natural Language Semantics 1*:181–234.

Chomsky, N. (1977). On Wh-Movement, in *Formal Syntax*, P. Culicover, T. Wasow, and A. Akmajian (eds.), 71-132. New York: Academic Press.

Chomsky, Noam (1973). Conditions on Transformations, in S. Anderson and P. Kiparsky, (eds.), *A Festschrift for Morris Halle*, pp. 232–286. Holt, Rinehart and Winstron, New York.

Chomsky, Noam (1981). *Lecture on Government and Binding*. Foris, Dordrecht.

Chomsky, Noam (1995). *The Minimalist Program*. Cambridge: MIT Press.

Desouvrey, Louis-Harry (2000). Romance Clitics and Feature Asymmetry: An Autosegmental-Based Approach. Doctoral dissertation, UQAM.

Desouvrey, Louis-Harry (2003). The Proper Treatment of Coreference Relations. www.semanticsarchive.net

Desouvrey, Louis-Harry (2008). Vector Effects on Wh-Extraction. www.lingBuzz.net

Desouvrey, Louis-Harry (2009). Nominal Predication, Relative Clauses, and *Wh*-Extraction: The Amazing Syntax of Haitian Creole Se. Lingbuzz.net.

Desouvrey, Louis-Harry (2018). The Syntax of Italian Clitics. Lingbuzz.net.

Desouvrey, Louis-Harry (2021). V2: The German Liberation Movement. www.lingBuzz.net.

Desouvrey, Louis-Harry (2022). Japanese Relative Clauses and the Parallelogram Law. *LingBuzz.net*.

Desouvrey, Louis-Harry (2024a). Feature Incompatibility in Russian: The Rise of the Genitive of Negation. lingBuzz.net.

Desouvrey, Louis-Harry (2024b). Anatomy of the Russian Agreement Mishmash. lingBuzz.net.

Desouvrey, Louis-Harry(2007), Wh-Interrogatives: The OCP Cycle. lingBuzz.net

Goldsmith, John (1976). Autosegmental Phonology. Doctoral dissertation , MIT.

Hayley Ross, Gennaro Chierchia, and Kathryn Davidson (2023). Quantifying weak and strong crossover for wh-crossover and proper names. In: Maria Onoeva, Anna Staňková, and Radek Šimík (eds.): *Proceedings of Sinn und Bedeutung 27*, pp. 535-553 Praha: Charles University.

Hornstein, Norbert (1995). *Logical Form: From GB to minimalism*. Oxford: Blackwell.

Ishii, Toru. 2006. A Nonuniform Analysis of Overt W*h*-Movement. *Linguistic Inquiry* 37:155–167.

Lasnik, Howard and Juan Uriagereka (1988). *A Course in GB Syntax*. Cambridge, MIT Press.

Lasnik, Howard and Tim Stowell (1991). Weakest Crossover. *Linguistic Inquiry*, 22: 687-720.

May, Robert (1995). *Logical Form*. Cambridge: MIT Press.

Poesio, Massimo, and Alessandro Zucchi (1992). On telescoping. In *Proceedings of SALT II*, ed. by Chris Barker and David Dowty, 347–366. Columbus: The Ohio State University, Department of Linguistics.

Postal, Paul (1971). Crossover Phenomena. New York: Holt, Rinehart and Winston.

Postal, Paul (1993). Remarks on Weak Crossover Effects. *Linguistic Inquiry* 24: 539-556.

Ross, Hayley, Gennaro Chierchia, and Kathryn Davidson (2023). Quantifying weak and strong crossover for wh-crossover and proper names. In: Maria Onoeva, Anna Staňková, and Radek Šimík (eds.): Proceedings of Sinn

und Bedeutung 27, pp. 535-553 Praha: Charles University.

Ruys, E. G. 2000. Weak crossover as a scope phenomenon. *Linguistic Inquiry* 31:513–539.

Safir, Ken (2015). Weak Crossover. lingbuzz.net

Shan, Chung-Chieh and Chris Barker (2006). Explaining Crossover and Superiority as Left-to-Right Evaluation. *Linguistics and Philosophy* 29:91–134.

Wasow, Thomas (1979. Anaphora in generative grammar. Ghent: E. Story-Scientia.